Reading

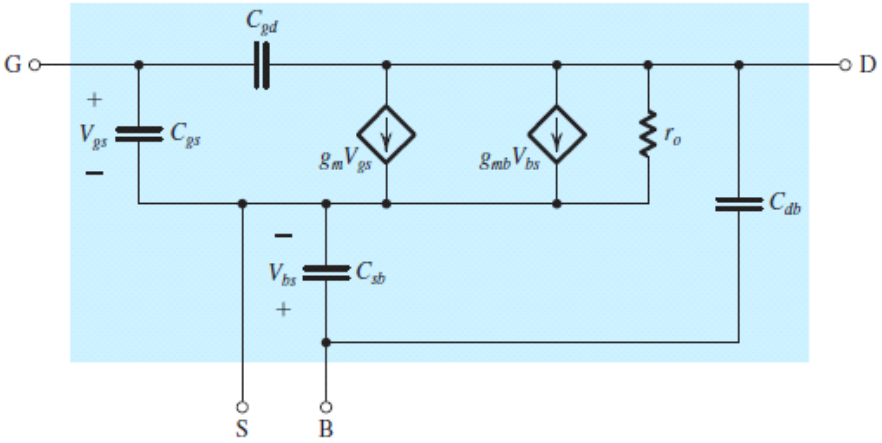
Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.2 High-Frequency Response of CS and CE Amplifier
- Chapter 10.3 The Method of Open-Circuit Time Constants
- Chapter 10.4 High-Frequency Response of Common-Gate and Cascode Amplifiers

Summary Table

Table 10.1 The MOSFET High-Frequency Model

Model



Model Parameters

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}| = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{|V_{OV}|}$$

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{|V_{SB}|}{V_0}}}$$

$$g_{mb} = \chi g_m, \quad \chi = 0.1 \text{ to } 0.2$$

$$r_o = |V_A|/I_D$$

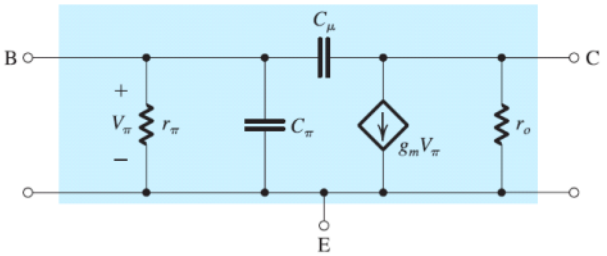
$$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{|V_{DB}|}{V_0}}}$$

$$C_{gs} = \frac{2}{3} WLC_{ox} + WL_{ov}C_{ox}$$

$$C_{gd} = WL_{ov}C_{ox}$$

$$f_T = \frac{g_m}{2\pi (C_{gs} + C_{gd})}$$

Table 10.2 The BJT High-Frequency Model

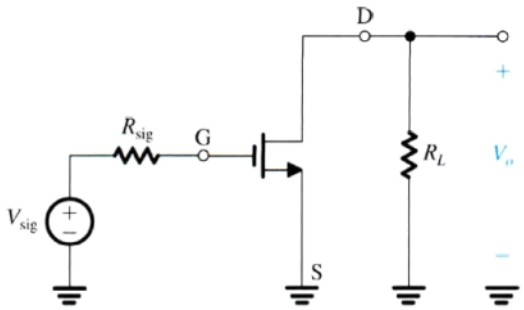


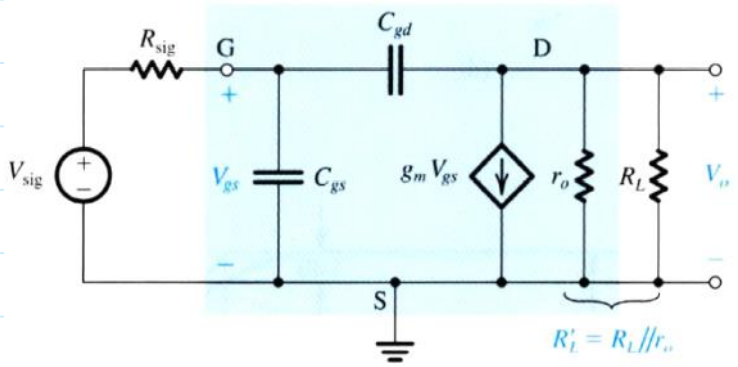
$$g_m = I_C/V_T \qquad r_o = |V_A|/I_C \qquad r_\pi = \beta_0/g_m$$

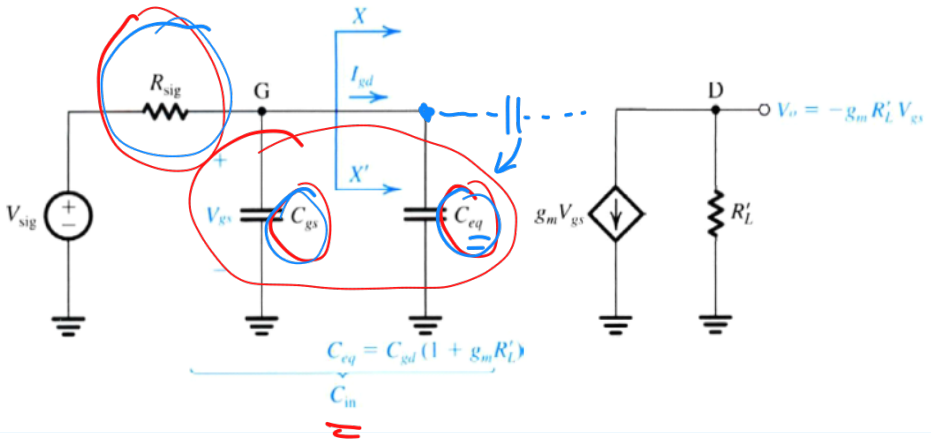
$$C_\pi + C_\mu = \frac{g_m}{2\pi f_T} \qquad C_\pi = C_{de} + C_{je} \qquad C_{de} = \tau_F g_m \qquad C_{je} \simeq 2C_{je0}$$

$$C_\mu = C_{jc0} \Big/ \left(1 + \frac{|V_{CB}|}{V_{0c}}\right)^m \qquad m = 0.3\text{–}0.5$$

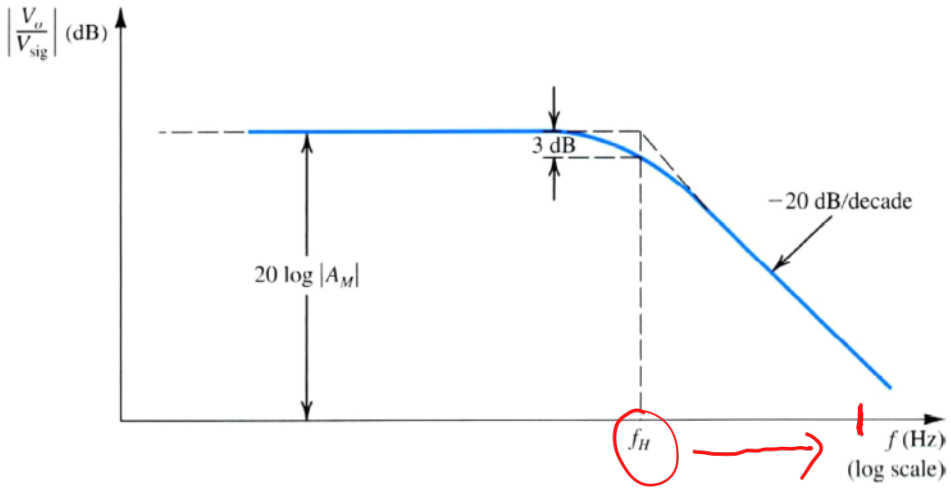
Recap







$$\frac{1}{2\pi\left(C_{gs} + C_{gd}(1+g_m R_L')R_{sig}\right)}$$

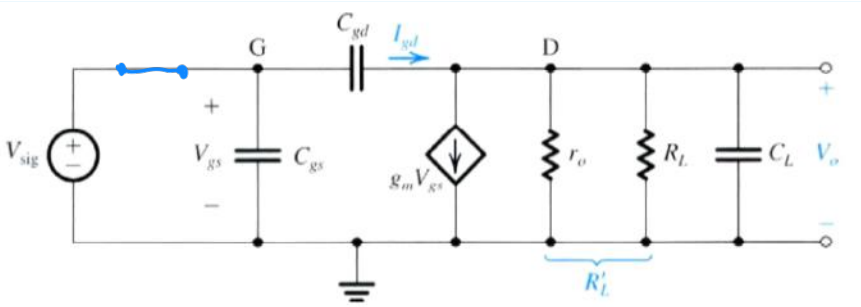


f_{Hn}

$$f_H = \frac{\omega_H}{2\pi} = \frac{1}{2\pi C_{in} R_{sig}}$$

$f_H \ll f_{Hn}$

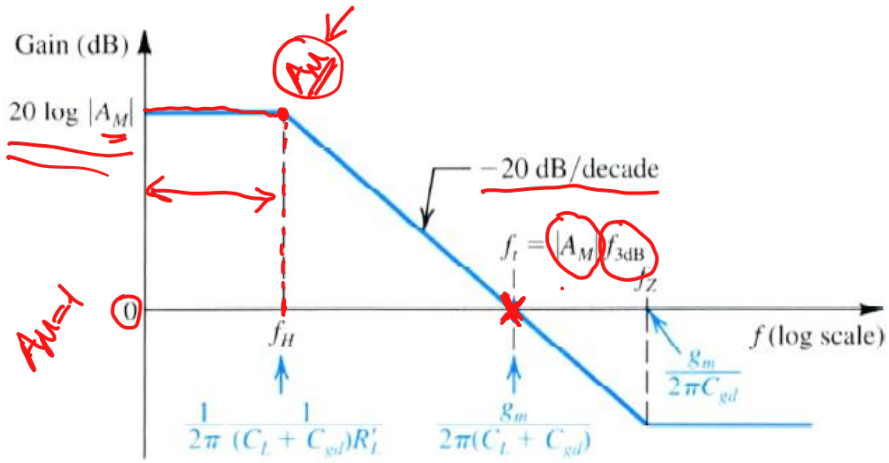
Recap



$$\frac{V_o}{V_{sig}} = \boxed{-g_m R'_L} \frac{1 - s(C_{gd}/g_m)}{1 + s(C_L + C_{gd})R'_L}$$

A_M

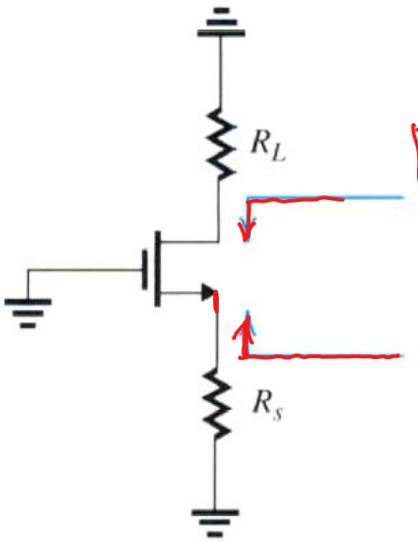
$$s_z = \frac{g_m}{C_{gd}} \qquad s_p$$



$$f_z = \frac{1}{2\pi} \frac{g_m}{C_{gd}}$$

$$\frac{1}{2\pi (C_L + C_{gd}) R'_L}$$

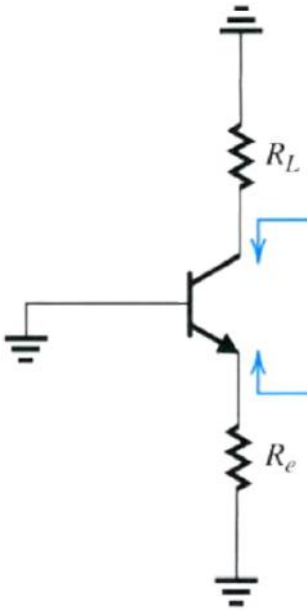
Recap



$$R_{out} = r_o + R_s + g_m r_o R_s$$
$$= r_o + (g_m r_o) R_s$$

$$R_{in} = \frac{r_o + R_L}{g_m r_o + 1}$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

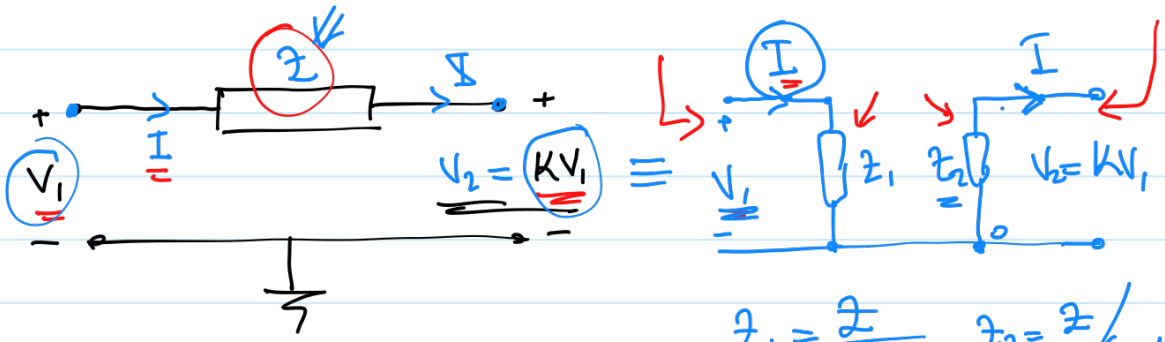
$A_{is} \simeq 1$



$$R_{out} = r_o + (R_e \| r_\pi) + g_m r_o (R_e \| r_\pi)$$
$$\simeq r_o + (g_m r_o)(R_e \| r_\pi)$$

$$R_{in} = r_\pi \| \left[\frac{r_o + R_L}{g_m r_o + 1} \right]$$
$$\simeq \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Miller's Theorem

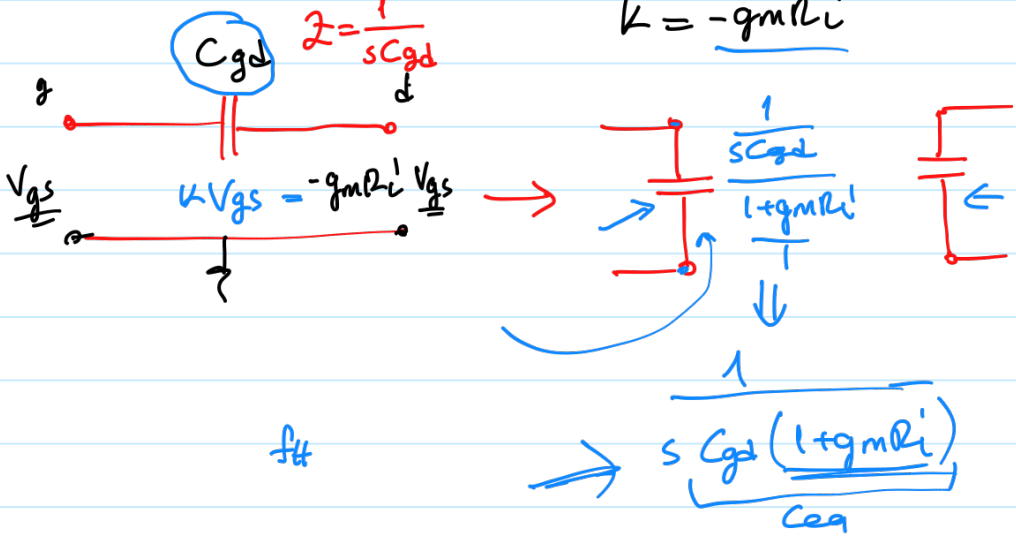


$$Z_1 = \frac{Z}{1-K} \qquad Z_2 = Z\left(1-\frac{1}{K}\right)$$

$$V_2 = KV_1$$

$$I = \frac{V_1 - V_2}{Z} = \frac{V_1 - KV_1}{Z} = \frac{V_1}{Z_1} \Rightarrow Z_1 = \frac{Z}{1-K}$$

$$I = \frac{V_1 - KV_1}{Z} = \frac{0 - V_2}{Z_2} = \frac{-KV_1}{Z_2}$$



$$f_t \rightarrow \frac{1}{s\,C_{gd}(1+g_mR_L')} \qquad C_{eq}$$

The Method of Open-Circuit Time Constants

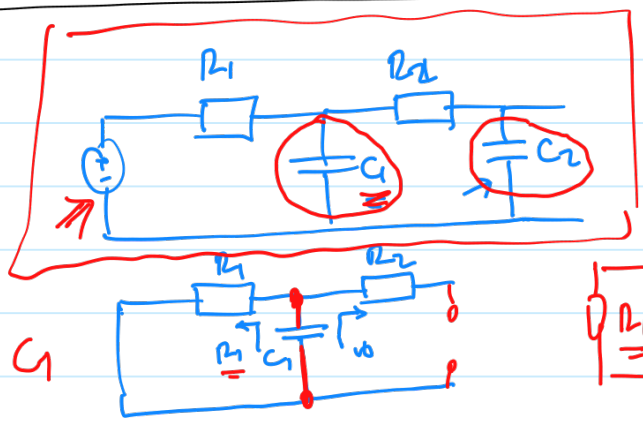
$$A(s) = A_M \cdot F_H(s)$$

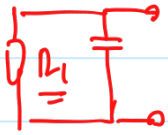
$$F_H(s) = \frac{(1+s/\omega_{z_1})(1+s/\omega_{z_2})\cdots(1+s/\omega_{z_n})}{(1+s/\omega_{p_1})(1+s/\omega_{p_2})\cdots(1+s/\omega_{p_n})}$$

$$F_H(s) \overset{n}{=} \frac{1}{1+s/\omega_{p_H}}$$

$$F(s) = \frac{1+a_1 s+a_2 s^2+\cdots+a_n s^n}{1+b_1 s+b_2 s^2+\cdots+b_n s^n}$$

$$b_1 = \sum_{i=1}^{n} C_i R_i$$

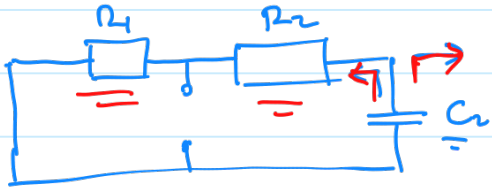


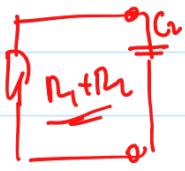


C_1

$$b_1 = \underbrace{R_1 C_1}_{\tau_1} + \underbrace{(R_1+R_2) C_2}_{\tau_2}$$

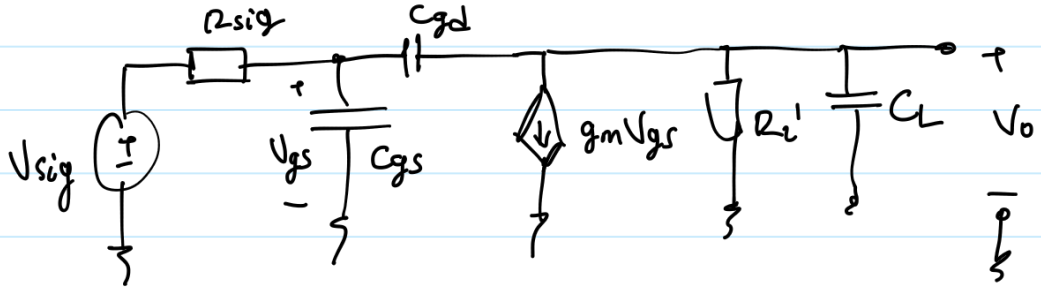
C_2

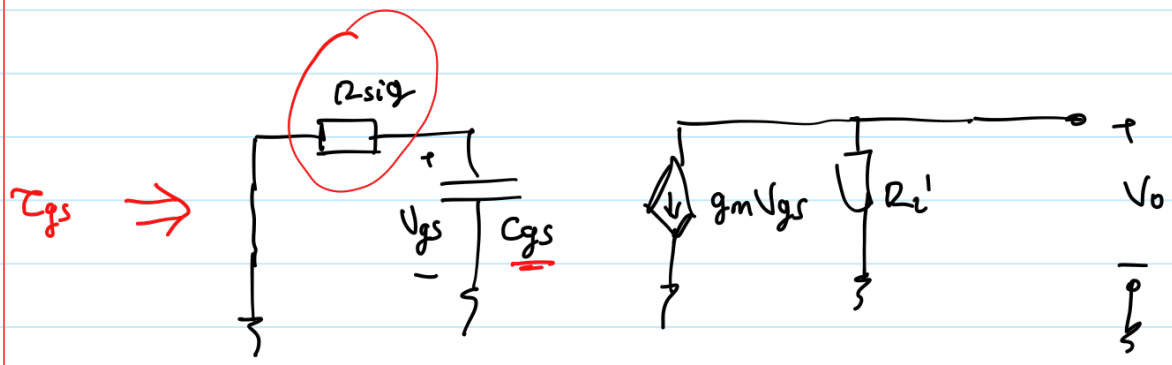




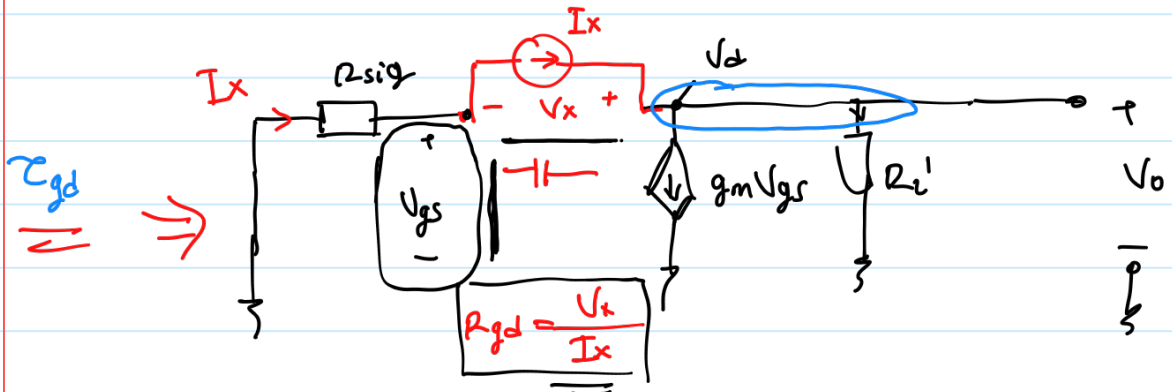
$$F(s) \overset{n}{=} \frac{1}{1+(\tau_1+\tau_2+\tau_3)s} = \frac{1}{1+\dfrac{s}{\omega_P}}, \quad \omega_P = \frac{1}{\tau_1+\tau_2+\tau_3}$$

Applying the Method of OCTC to the CS Amplifiers





$$R_{gs} = R_{sig} \qquad \tau_{gs} = R_{sig} \cdot C_{gs}$$



$$V_{gs} = -I_x R_{sig}$$

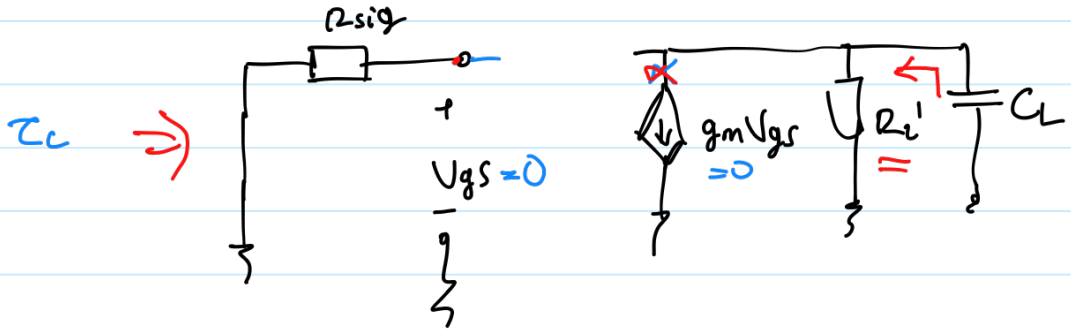
$$V_d = V_x + V_{gs}$$

$$I_x = g_m V_{gs} + \frac{V_d}{R_L'} \qquad I_x = g_m V_{gs} + \frac{V_x + V_{gs}}{R_L'}$$

$$I_x = g_m\left(-I_x R_{sig}\right) + \frac{V_x}{R_L'} + \frac{\left(-I_x R_{sig}\right)}{R_L'}$$

$$R_{gd} = \frac{V_x}{I_x} = R_{sig}\left(1 + g_m R_L'\right) + R_L' \qquad \tau_{gd} = R_{gd} \cdot C_{gd}$$

Example



$\tau_L \Rightarrow \quad \tau_L = R_L' \cdot C_L$

$$\Rightarrow b_1 = \sum_{i=1}^{n} R_i C_i = C_{gs} R_{gs} + C_{gd} R_{gd} + C_L R_L$$

$$\tau_H = b_1 = C_{gs} R_{sig} + C_{gd}\left(R_{sig}(1+g_m R_i') + R_i'\right) + C_L R_L'$$

$$\omega_H = \frac{1}{\tau_H} \Rightarrow f_H = \frac{1}{2\pi \tau_H}$$

$$f_H = \frac{1}{2\pi\left[C_{gs} + C_{gd}(1+g_m R_i')\right] R_{sig} + (C_{gd} + C_L) R_L'}$$

$$\frac{1}{2\pi\left(C_{gs} + C_{gd}(1+g_m R_i')\right) R_{sig}}$$

R_{sig} is large

$$\frac{1}{2\pi (C_L + C_{gd}) R_L'}$$

R_{sig} is small

Template Ex. 10.5

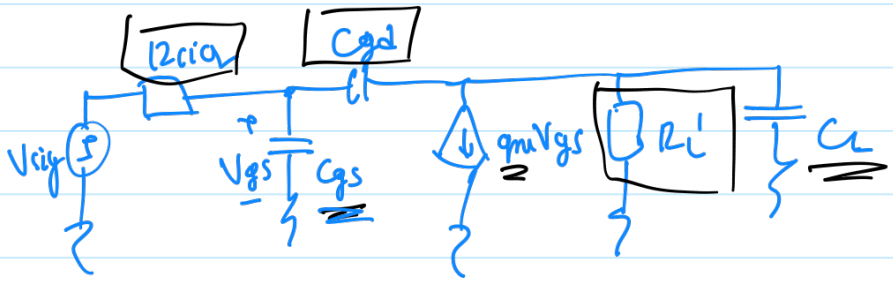
$\Rightarrow$ $g_m = 2mA/V$ $C_{gs} = 20fF$ $C_{gd} = 5fF$ $C_L = 25fF$

$R_{sig} = 20k\Omega$ $R_L' = 10k\Omega$

$H(s) = \dfrac{(1 - s/\omega_{z_1})(\cdots)}{1 + b_1 s \cdots}$ $s_z = \omega_{z_1}$

f_H

$\Rightarrow$



$\tau_H = \tau_{gs} + \tau_{gd} + \tau_{CL}$ $\Rightarrow$ $f_H = \dfrac{1}{2\pi \tau_H}$

$R_{gs} = R_{sig} = 20k\Omega$

$R_{gd} = R_{sig}(1 + g_m R_L') + R_L' = 430k\Omega$

$R_{CL} = R_L' = 10k\Omega$

$f \Rightarrow 10^{-15}$
$p \Rightarrow 10^{-12}$

$\tau_{gs} = C_{gs} R_{gs} = 20k \cdot 20 \times 10^{-15} = 400ps$

$\tau_{gd} = C_{gd} R_{gd} = 5 \times 10^{-15} \times 430k = 2150ps$

$\tau_{CL} = C_L \cdot R_{CL} = 25 \times 10^{-15} \times 10k = 250ps$

$\tau_H = \tau_{gs} + \tau_{gd} + \tau_{CL} = (400 + 2150 + 250)ps$

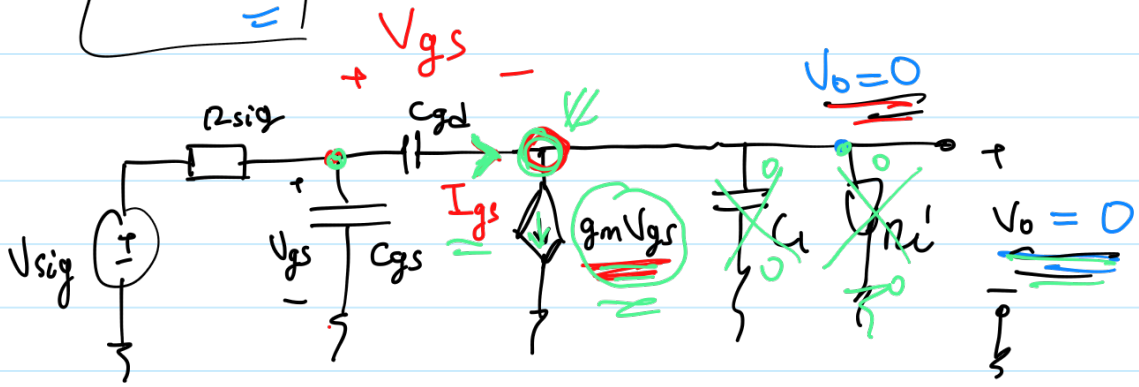
$= 2800ps$

$f_H = \dfrac{1}{2\pi \tau_H} = \dfrac{1}{2\pi \cdot 2800ps} \simeq 56.8MHz$

$H(s_z) = 0$ $\dfrac{V_o}{V_{sig}} = H(s)$ $\Rightarrow$ $V_o = 0$ @ $s = s_z$

Template

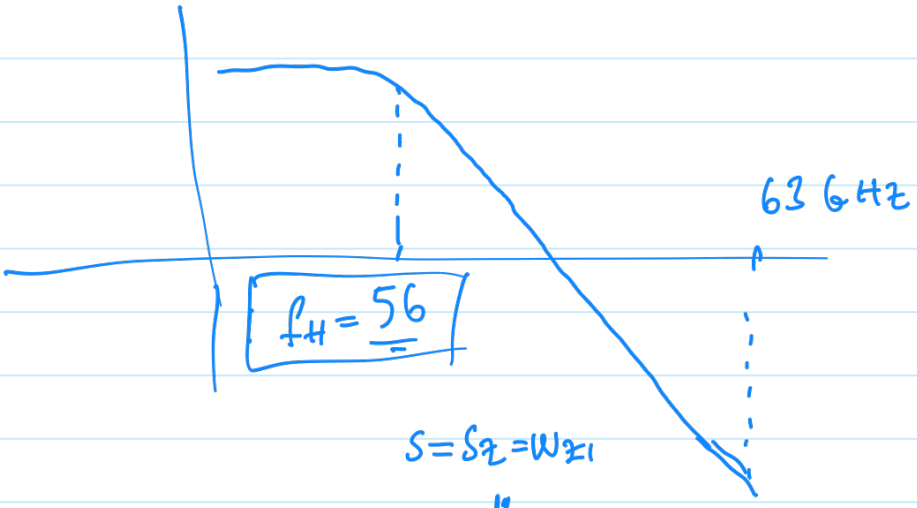
$s \to s_z$



$$I_{gs} = s_z V_{gs} \cdot C_{gd} = g_m V_{gs}$$

$$s_z C_{gd} = g_m \Rightarrow s_z = \frac{g_m}{C_{gd}}$$

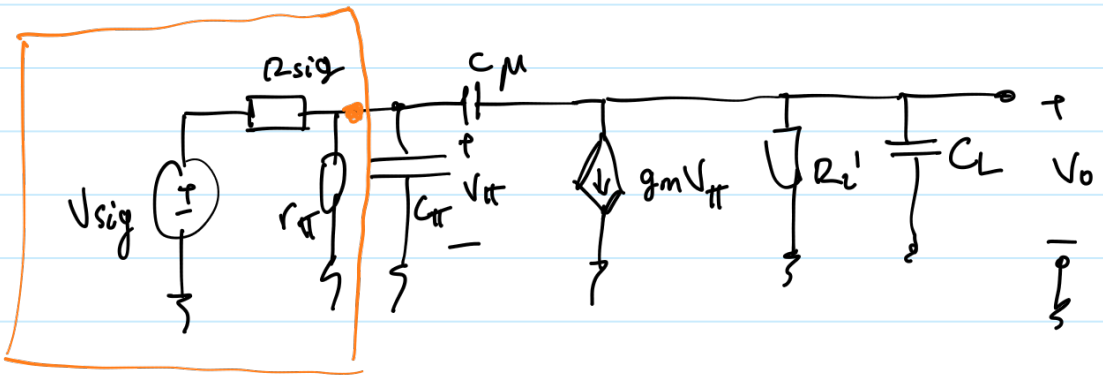
$$f_z = \frac{g_m}{2\pi C_{gd}} = \frac{2m}{2\pi \cdot 5\times10^{-15}} = 63.7\,GHz$$

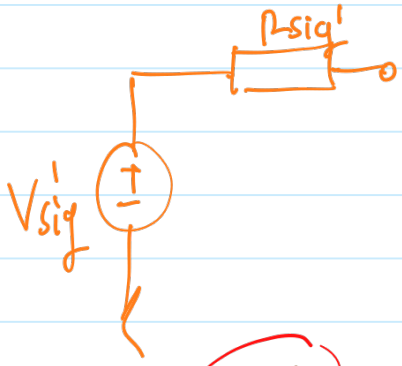


$$\frac{V_o}{V_{sig}} = H(s_z) = \frac{(1 - s_z/\omega_{z1}) \cdots}{(1 + s_z/\omega_{p1})(\cdots)} = 0$$

$f_H \Rightarrow V_o(s_z) = 0$

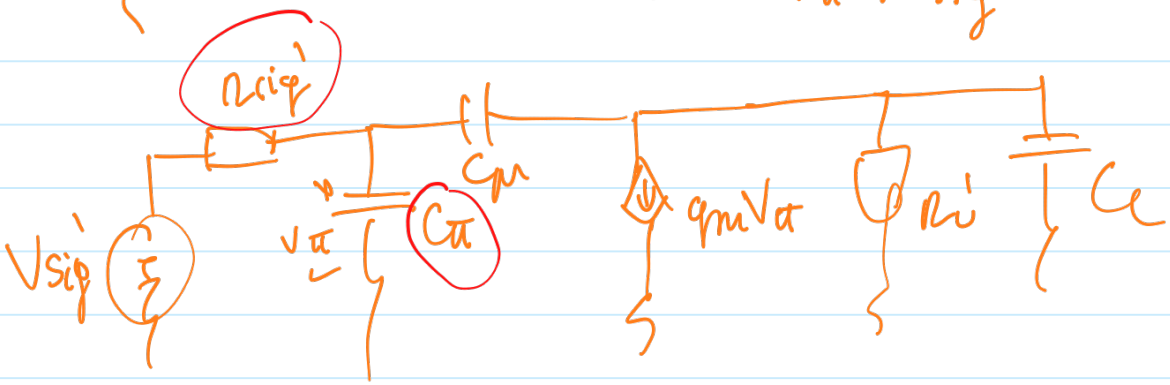
Applying the Method of OCTC to the CE Amplifiers

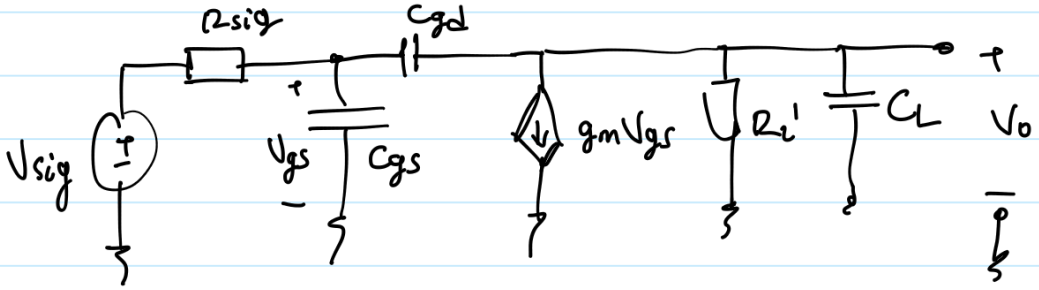




$$R_{sig}' = R_{sig} \,//\, r_\pi$$

$$V_{sig}' = \frac{r_\pi}{r_\pi + R_{sig}} \cdot V_{sig}$$





High Frequency Response of Common Gate Amplifier

Example